How Often Should You Recalibrate? A Closed-Form Cadence Law for Deployed Probabilistic Forecasters

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***Abstract*—Deployed probabilistic forecasters lose calibration over time, and every practical remedy — refitting a Platt or isotonic map, re-conformalizing intervals, retraining — costs something. Existing work treats the timing question algorithmically: drift detectors, Bayesian monitors, and empirical frequency studies decide *reactively* when to update. We instead derive the optimal update cadence in closed form. Modeling the post-recalibration calibration gap as $g(t) = \gamma t^{\beta}$ and bridging gap to loss through the reliability term of the Brier decomposition (Lemma 1: $\mathrm{REL} \geq \mathrm{ECE}^2$), renewal–reward optimization yields a one-line law: $T^* = \left[(2\beta+1)\, c \,/\, (2\beta\rho\gamma^2)\right]^{1/(2\beta+1)}$, where c is the recalibration cost and ρ the forecast rate. Linear drift (β=1) gives a cube-root law, diffusive drift (β=$\frac{1}{2}$) recovers an economic-order-quantity square-root law, and the loss penalty for operating at xT^* is the closed-form $R(x) = \frac{2\beta}{2\beta+1}x^{-1} + \frac{x^{2\beta}}{2\beta+1}$, so a $2\times$ error in the cost estimate inflates loss by only 5.8% at β=1. Monte-Carlo experiments confirm the law (mean $\hat{T}/T^* = 0.993$ over an 18-point parameter grid) and its scaling exponents, and quantify the monitoring sample size needed for reliable plug-in use (≈ 200 resolved outcomes per staleness bucket). Finally, we measure the law's central primitive on a live system: across 564 resolved interval forecasts for six National Stock Exchange of India equities, the nominal-90% conformal band under-covers by 14.8 percentage points immediately after fitting and decays at $\hat{\gamma} = 0.80 \pm 0.67$ pp/day, an estimate independently corroborated by the system's probability-calibration telemetry (0.76 pp/day). Under the measured drift the law prescribes recalibration every 2.8–6.2 days rather than at the monthly cadences common in practice.**

***Index Terms*—calibration drift, recalibration, Brier score, expected calibration error, renewal–reward, conformal prediction, MLOps, financial forecasting**

I. Introduction

A probability that meant 60% on the day a model was fit rarely still means 60% a month later. Calibration — the property that events forecast with probability p occur with frequency p — is the first casualty of distribution shift, decaying faster than discrimination in deployed clinical models [11], [12] and, as we measure below, in financial forecasting systems within *days*. The standard remedies are cheap relative to retraining: refit a Platt scaler [7], an isotonic map [8], a temperature parameter [5], or re-conformalize an interval predictor [9], [10]. Yet none of the literature answers the operational question every deployment faces: *how often*?

Current practice picks cadences by convention (nightly, monthly, quarterly), and current research picks them *reactively*: drift detectors that trigger updates [12], cost-aware decision algorithms such as CARA [13], Bayesian "learning debt" monitors [14], anytime-valid calibration tests [15], and empirical studies of retraining frequency in retail demand forecasting [16]. These approaches are valuable, but they are procedures, not laws: they output a decision, not an interpretable formula exposing how the optimum moves with cost and drift. The analogous question in operations research — replace a part now or later — was answered a century ago by closed-form policies: the economic order quantity of Harris [17] and the age-replacement policies of reliability theory [18]. To our knowledge no equivalent closed form exists for probability recalibration.

This paper supplies one. The derivation has three ingredients, each standard in isolation; the contribution is the bridge between them and the resulting law. First, the Murphy decomposition [2] isolates the avoidable cost of miscalibration as the reliability term REL of the Brier score [1], the quantity a perfect recalibration map removes. Second, an elementary inequality (Lemma 1) bounds REL below by the square of the expected calibration error (ECE) [4], turning any *measured* calibration-gap trajectory $g(t)$ into a loss trajectory $g(t)^2$. Third, renewal–reward optimization over the recalibration cycle converts the loss trajectory and a per-event recalibration cost c into a closed-form optimal cadence.

Contributions.

1) *A cadence law* (Theorem 1). For calibration gap $g(t) = \gamma t^{\beta}$ after each recalibration, forecast rate ρ, and recalibration cost c, the loss-minimizing interval is

$$T^* = \left[\frac{(2\beta+1)\, c}{2\beta\,\rho\,\gamma^2}\right]^{\frac{1}{2\beta+1}},$$

a one-parameter *family* of laws: cube-root under linear drift, square root (the EOQ form [17]) under diffusive drift.

2) *Operational corollaries in closed form*: the optimal loss rate $A^* = \frac{2\beta+1}{2\beta}\, c/T^*$; scaling elasticities $\partial \log T^*/\partial \log c = 1/(2\beta+1)$ and $\partial \log T^*/\partial \log \gamma = -2/(2\beta+1)$; and a misspecification penalty curve $R(x)$ showing the policy is forgiving to errors in c (attenuated to the $1/(2\beta+1)$ power) but less so to errors in γ.

3) *Monte-Carlo validation*: across an 18-cell grid in (β, c, γ) the empirical optimum matches T^* with mean

ratio 0.993; measured scaling exponents agree with theory; the law remains accurate (within 1–18%) when the drift mechanism violates the uniform-gap assumption; and plug-in use requires ≈ 200 resolved outcomes per staleness bucket before the estimate of γ is decision-grade.

4) *A live measurement of the primitive*. Using a deployed Indian equity advisory system whose model is re-fit at every forecast issuance — so that forecast lead time *equals* model staleness — we measure interval-calibration decay on 564 resolved forecasts: a 14.8 pp baseline under-coverage at $t{=}1$ and $\hat{\gamma} = 0.80 \pm 0.67$ pp/day of further decay, corroborated at 0.76 pp/day by an independent probability-ECE telemetry channel of the same system. The law then yields concrete cadences (2.8–6.2 days) for measured costs.

Scope is stated plainly: the law optimizes *recalibration* (an output-map refresh that restores reliability), not full retraining (which also restores discrimination); the empirical drift estimates carry the wide confidence intervals of a three-month, six-ticker ledger; and the drift exponent β is weakly identified by our data ($\hat{\beta} = 0.23 \pm 0.22$), which is precisely why the law is stated and validated as a family over β rather than a single exponent.

II. Related Work

Calibration and its measurement. Proper scoring rules [1], [3] and the Murphy decomposition [2] underpin our loss accounting; ECE in its ℓ_1 and ℓ_2 forms [4], [6] supplies the measurable gap. Modern recalibration maps — Platt [7], isotonic [8], temperature [5] — are the cheap maintenance actions whose cadence we optimize; conformal methods [9] and their adaptive variants [10] play the same role for intervals. Adaptive conformal inference [10] in particular performs continuous small corrections; our law addresses the complementary regime of discrete recalibration events with non-negligible cost.

Calibration drift. That deployed models lose calibration first is documented extensively in clinical prediction [11], with data-driven updating strategies proposed in [12]. These works detect and correct drift but do not derive an optimal cadence.

Retraining economics. CARA [13] optimizes retrain/keep decisions over data streams; the posterior "learning debt" framework [14] triggers retraining when calibrated expected regret exceeds cost; [16] studies retraining frequency for retail forecasting empirically; [15] gives anytime-valid monitoring tests. All are sequential or empirical procedures. None provides a closed-form interval, scaling exponents, or a misspecification penalty in analytic form. Conversely, classical maintenance theory [17]–[19] has exactly these closed forms but has not, to our knowledge, been connected to probability calibration via the Brier reliability bridge.

Positioning. Our law is to CARA-style monitors what EOQ is to just-in-time controllers: not a replacement but a design rule — an interpretable first-order answer that exposes the relevant elasticities, costs nothing to evaluate, and degrades gracefully (Corollary 3) when its inputs are rough.

III. Preliminaries

Binary outcomes $y_i \in \{0,1\}$ are forecast with probabilities $p_i \in [0,1]$ at rate ρ forecasts per day. The Brier score $\mathrm{BS} = \frac{1}{n}\sum_i (p_i - y_i)^2$ [1] admits the Murphy decomposition [2]

$$\mathrm{BS} \;=\; \underbrace{\sum_b w_b\,(\bar{p}_b - \bar{y}_b)^2}_{\mathrm{REL}} \;-\; \mathrm{RES} \;+\; \mathrm{UNC}, \qquad (1)$$

over forecast bins b with mass w_b, mean forecast $\bar{p}_b$ and event frequency $\bar{y}_b$. REL is the *reliability* (calibration) penalty; RES (resolution) and UNC (outcome entropy) are unchanged by any monotone relabeling of the forecasts. Applying the conditional-event-frequency map $\kappa(p) = \mathbb{E}[y \mid p]$ — the ideal recalibration, approximated in practice by Platt/isotonic/temperature fitting — removes REL and only REL. Hence REL is exactly the per-forecast loss a recalibration event recovers, and we call it the *miscalibration loss rate* when expressed per day (multiplying by ρ).

The binned ℓ_1 and ℓ_2 expected calibration errors are $\mathrm{ECE}_1 = \sum_b w_b\,|\bar{p}_b - \bar{y}_b|$ and $\mathrm{ECE}_2 = (\sum_b w_b(\bar{p}_b - \bar{y}_b)^2)^{1/2}$ [4], [6]; note $\mathrm{ECE}_2^2 = \mathrm{REL}$ by definition. For interval forecasters with nominal level $1-\alpha$, the analogous gap is $|(1-\alpha) - \mathrm{cov}(t)|$, the deviation of empirical coverage from nominal — the form we measure on the live ledger in Section VI.

Renewal–reward. If a system regenerates at every recalibration (cycle length T, cycle cost = recalibration cost + accumulated drift loss), the long-run average cost rate equals expected cycle cost over expected cycle length [19], reducing cadence choice to a one-dimensional minimization.

IV. The Recalibration Cadence Law

A. Assumptions

Assumption 1 (Power-law gap growth): Let t be the time since the last recalibration. The ℓ_2 calibration gap follows $g(t) = \gamma\, t^{\beta}$ with drift rate $\gamma > 0$ and exponent $\beta > 0$, and recalibration resets g to (approximately) zero.

Assumption 2 (Drift hits calibration, not discrimination): Over the horizon between recalibrations, RES and UNC in (1) are unaffected by the drift; only REL grows.

Assumption 3 (Stationary economics): Each recalibration costs $c > 0$, expressed in the same units as Brier loss accumulated per day (Section VII discusses this commensuration), and ρ, γ, β, c are constant across cycles.

Assumption 1 nests the two mechanisms one expects *a priori*: a deterministic environment trend gives $\beta = 1$; a gap that diffuses (variance linear in time) gives $\beta = \frac{1}{2}$ in root-mean-square. Assumption 2 is the standard recalibration premise — it is what Platt/isotonic refresh can and cannot fix — and we report evidence of its partial violation in our own data (discrimination also decays; Fig. 4b) together with the consequence: when discrimination decays too, T^* bounds the *recalibration* cadence and full retraining is a separate, slower control loop.

B. The gap–loss bridge

Lemma 1 (Calibration gap lower-bounds recoverable loss): For any binning with weights w_b summing to one, $\mathrm{REL} \geq \mathrm{ECE}_1^2$, with equality iff the absolute gap $|\bar{p}_b - \bar{y}_b|$ is constant across bins; and $\mathrm{REL} = \mathrm{ECE}_2^2$ identically.

Proof: By Cauchy–Schwarz with weights, $(\sum_b w_b|g_b|)^2 \leq (\sum_b w_b)(\sum_b w_b g_b^2) = \mathrm{REL}$, where $g_b = \bar{p}_b - \bar{y}_b$. Equality holds iff $|g_b|$ is w-a.s. constant. The ℓ_2 identity is definitional. ∎

The lemma is elementary, but it is the hinge of the method: practitioners measure ECE, while the loss that recalibration recovers is REL. Stating the drift model in ℓ_2 (γ an ECE_2 rate) makes everything below exact; stating it in the more commonly logged ℓ_1 makes the predicted loss a *lower* bound, hence T^* computed from ℓ_1 telemetry is an *upper* bound on the true optimal interval: recalibrate *at least* that often.

C. The law

Theorem 1 (Recalibration cadence law): Under Assumptions 1–3, the long-run average loss rate of the policy "recalibrate every T days" is

$$A(T) \;=\; \frac{c}{T} \;+\; \frac{\rho\,\gamma^2\,T^{2\beta}}{2\beta+1}, \qquad (2)$$

which has the unique global minimizer

$$\boxed{T^* \;=\; \left[\frac{(2\beta+1)\,c}{2\beta\,\rho\,\gamma^2}\right]^{\frac{1}{2\beta+1}}} \qquad (3)$$

with optimal loss rate $A^* = A(T^*) = \frac{2\beta+1}{2\beta} \cdot \frac{c}{T^*}$.

Proof: Within a cycle the per-day recoverable loss at staleness t is $\rho\,\mathrm{REL}(t) = \rho\,g(t)^2 = \rho\gamma^2 t^{2\beta}$ (Lemma 1, ℓ_2 form; Assumption 2 keeps the remaining Brier terms common to all policies, so they cancel from the comparison). Cycle cost is $c + \rho\gamma^2 \int_0^T t^{2\beta}dt = c + \rho\gamma^2 T^{2\beta+1}/(2\beta+1)$; dividing by T gives (2) by renewal–reward [19]. Setting $A'(T) = 0$ yields $c = \frac{2\beta}{2\beta+1}\rho\gamma^2 T^{2\beta+1}$, whose right side is strictly increasing in T for any $\beta > 0$; hence the stationary point is unique, with $A' < 0$ before and $A' > 0$ after it. Substituting back gives $\rho\gamma^2(T^*)^{2\beta}/(2\beta+1) = c/(2\beta T^*)$ and thus $A^* = c/T^* + c/(2\beta T^*)$. ∎

Corollary 1 (Canonical regimes): Linear drift ($\beta = 1$): $T^* = \left(3c/2\rho\gamma^2\right)^{1/3}$ and $A^* = \frac{3}{2}\,c/T^*$ — a *cube-root law*. Diffusive drift ($\beta = \frac{1}{2}$): $T^* = \left(2c/\rho\gamma^2\right)^{1/2}$ — exactly the square-root form of the economic order quantity [17], with calibration variance playing the role of holding cost.

Corollary 2 (Scaling elasticities): $\dfrac{\partial \log T^*}{\partial \log c} = \dfrac{1}{2\beta+1}$, $\dfrac{\partial \log T^*}{\partial \log \gamma} = -\dfrac{2}{2\beta+1}$, and $A^* \propto c^{\frac{2\beta}{2\beta+1}}\gamma^{\frac{2}{2\beta+1}}$. At $\beta = 1$: doubling cost stretches the cadence by $2^{1/3} \approx 1.26$; doubling drift shrinks it by $2^{-2/3} \approx 0.63$; and the best achievable loss grows as $(c\gamma)^{2/3}$.

Corollary 3 (Misspecification penalty): Operating at $T = xT^*$ costs

$$R(x) \;=\; \frac{A(xT^*)}{A^*} \;=\; \frac{2\beta}{2\beta+1}\,\frac{1}{x} \;+\; \frac{x^{2\beta}}{2\beta+1}. \qquad (4)$$

Errors in c enter x through the $1/(2\beta{+}1)$ power (Corollary 2); at $\beta{=}1$ a $2\times$ misestimate of c gives $x = 2^{\pm 1/3}$ and $R = 1.058$ — a 5.8% loss penalty — while a $2\times$ misestimate of γ gives $x = 2^{\mp 2/3}$ and $R \leq 1.26$.

D. Imperfect recalibration and other extensions

Theorem 2 (Affine gap, $\beta = 1$): If recalibration is imperfect or the environment shifts at fit time, so that $g(t) = E_0 + \gamma t$ with residual gap $E_0 \geq 0$, then $A(T) = c/T + \rho\,(E_0^2 + E_0\gamma T + \gamma^2 T^2/3)$ and T^* is the unique positive root of

$$\tfrac{2}{3}\rho\gamma^2 T^3 + \rho E_0 \gamma T^2 - c \;=\; 0, \qquad (5)$$

solvable by Cardano's formula. T^* is decreasing in E_0: residual miscalibration makes *more* frequent recalibration optimal (the cross-term $E_0\gamma t$ is recoverable even though E_0^2 is not), and (5) reduces to the cube-root law as $E_0 \to 0$.

Proof: Expand $\int_0^T (E_0 + \gamma s)^2 ds = E_0^2 T + E_0\gamma T^2 + \gamma^2 T^3/3$, apply renewal–reward, drop the T-free term ρE_0^2, differentiate. The left side of (5) is strictly increasing in $T > 0$, giving uniqueness; monotonicity in E_0 follows from the implicit function theorem since $\partial/\partial E_0$ of the left side is $\rho\gamma T^2 > 0$. ∎

Remark 1: Exponential drift $g(t) = \gamma(e^{\lambda t} - 1)/\lambda$ admits the same treatment with a transcendental first-order condition; its small-λT expansion recovers (3) with $\beta = 1$. We omit it: over horizons of days to weeks the power family already spans the empirically plausible curvatures.

E. Practical recipe

The law is operational in four steps, all from logs a calibrated deployment keeps anyway: (1) after each recalibration, bucket resolved outcomes by staleness t and record the gap $\hat{g}(t)$ (ℓ_1 ECE, ℓ_2 ECE, or coverage gap); (2) fit $\hat{\gamma}$, and $\hat{\beta}$ if the data can bear it, otherwise fix β by mechanism knowledge (Corollary 1) and report both endpoints; (3) price c in Brier-equivalent units — the loss you would accept to avoid one refresh (compute + validation + operational risk); (4) apply (3); Corollary 3 bounds the cost of the remaining uncertainty. Section V (E4) quantifies step (2): ≈ 200 resolved outcomes per staleness bucket suffice; below ≈ 50 the plug-in estimate is not decision-grade and $\hat{\gamma}$ should be floored or pooled across models.

V. Simulation Validation

All experiments use a logit-normal ground-truth process ($q_t = \sigma(z)$, $z \sim \mathcal{N}(0.2, 0.9^2)$), Bernoulli outcomes, Brier loss measured against the perfectly calibrated benchmark, and the recorded seed; code accompanies the paper. Drift is imposed as a uniform additive gap $p = \mathrm{clip}(q + \gamma t^\beta)$ except where stated.

E1 — the law locates the optimum. For $\beta \in \{0.5, 1\}$, $c \in \{0.002, 0.01, 0.05\}$, $\gamma \in \{0.004, 0.008, 0.016\}$ (18 cells,

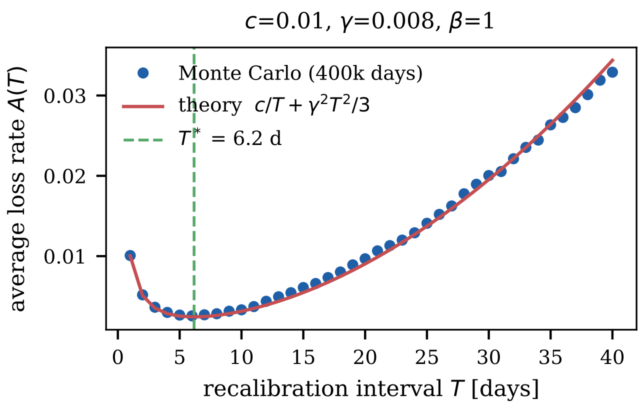

Fig. 1. Average daily loss versus recalibration interval: Monte-Carlo (dots) against the analytic $A(T)$ of (2) (line); the dashed vertical line is the closed-form T^*.

400,000 simulated days each) we sweep T on a geometric grid and compare the empirical argmin $\hat{T}$ with T^* from (3). The mean ratio $\hat{T}/T^*$ is 0.993; the worst cell deviates by $|\log(\hat{T}/T^*)| = 0.33$, occurring (as (4) predicts) where the minimum is flattest, so the loss consequence of those deviations is small. Fig. 1 shows a representative loss curve: Monte-Carlo points sit on the analytic $A(T)$ with the minimum at $T^* = 6.2$ days.

E2 — scaling exponents. Regressing $\log \hat{T}$ on $\log c$ over $c \in [10^{-3}, 10^{-1}]$ yields slopes 0.357 (β=1; theory 1/3) and 0.445 (β=0.5; theory 1/2); against $\log \gamma$, slopes -0.772 (theory $-2/3$) and -1.085 (theory -1). The residual bias is the integer-day discretization of $\hat{T}$. Fig. 2 displays both panels.

E3 — mechanism robustness. Real drift is not a uniform shift; e.g. temperature-style drift ($\operatorname{logit} p = (1+kt) \operatorname{logit} q$) concentrates the gap at extreme forecasts, making $\mathrm{REL} > \mathrm{ECE}_1^2$ strictly (Lemma 1). Imposing temperature drift whose ℓ_1 gap matches γt, the empirical optimum stays within 1–18% of T^* computed from either the ℓ_1 rate or the measured ℓ_2 rate — inside the flat region of (4) where the loss penalty is under 2%.

E4 — how much monitoring data does plug-in use need? We estimate $\hat{\gamma}$ from a single 15-day post-recalibration window with m resolved outcomes per day, apply (3), and price the resulting cadence with (4) (400 replications). Table I reports excess loss versus the oracle cadence: with m=200 the plug-in policy is within 5.3% of oracle on average; with m=50 the slope estimate is noise-dominated and the policy unusable without regularization — a concrete minimum-telemetry requirement for cadence-by-formula.

VI. Measuring the Primitive on a Live System

A. System and identification strategy

The measurement platform is a deployed retail equity advisory system for National Stock Exchange of India (NSE) tickers. At every forecast issuance the system re-fits its model stack on the trailing window and emits, for each horizon

TABLE I
E4: Plug-in cadence from noisy $\hat{\gamma}$ (β=1, c=0.01, γ=0.008, 400 reps).

outcomes/day m	mean excess loss	90th pct.
50	+608.6%	+438.8%
200	+5.3%	+13.1%
1000	+0.5%	+1.3%

day $k \in \{1, \ldots, 10\}$ (plus a 30-day swing channel), a point forecast, a directional call, and a split-conformal 90% interval; every forecast is written to an append-only ledger and resolved against realized prices. Because the stack is re-fit at issuance, *forecast lead time equals model staleness*: a k-day-ahead forecast is exactly a forecast evaluated k days after fitting. The ledger therefore identifies the gap trajectory $g(t)$ without any instrumentation beyond what the system already logs — with the honest caveat that lead-time difficulty and staleness are conflated by this design, a limitation we return to in Section VII.

We analyze the complete ledger snapshot of 2026-06-12: 564 fully resolved interval forecasts (issued 2026-04-19 to 2026-06-11) across six liquid NSE names (HDFCBANK, ICICIBANK, INFY, RELIANCE, SBIN, TCS), nominal coverage 90%, staleness support concentrated on $t \in [1, 12]$ calendar days with $n \geq 38$ per bucket.

B. Results

Fig. 4a shows the coverage gap by staleness. Three findings:

(i) Baseline miscalibration is immediate and large. At t=1 the nominal-90% band covers only 71.1% (n=38): $E_0 \approx 14.8$ pp of under-coverage exists *on the day after fitting*. The mechanism is visible in the sample period — bands fit on a trailing window that included a calmer regime, deployed into a high-volatility drawdown. By Theorem 2, large E_0 *tightens* the optimal cadence.

(ii) The gap drifts upward with staleness. The affine fit gives $\hat{\gamma} = 0.80 \pm 0.67$ pp/day (p=0.26; coverage 66% at t=5, 55% at t=9). The estimate is individually noisy — exactly the E4 regime with $m \approx 45$ per bucket — but it is corroborated by an independent channel of the same system: probability-calibration telemetry records holdout ECE of 2.96% at fit time against 25.67% over the live 30-day resolution window, an implied drift of 0.76 pp/day. Two channels (interval coverage; probability ECE), two estimators, one magnitude: $\gamma \approx 0.8$ pp/day.

(iii) The drift exponent is weakly identified. The free-exponent fit gives $\hat{\beta} = 0.23 \pm 0.22$; with the affine intercept absorbed, $\beta \in \{\frac{1}{2}, 1\}$ are both compatible. This is why Theorem 1 is stated as a family and why the cadence below is reported as a range over β, not a point.

Fig. 4b shows the companion effect on discrimination: the directional hit-rate decays from 59–69% at $t \in [3, 4]$ toward 46–48% at $t \geq 10$ (-1.11 ± 0.68 pp/day, p=0.13), i.e. Assumption 2 is conservatively violated — staleness costs

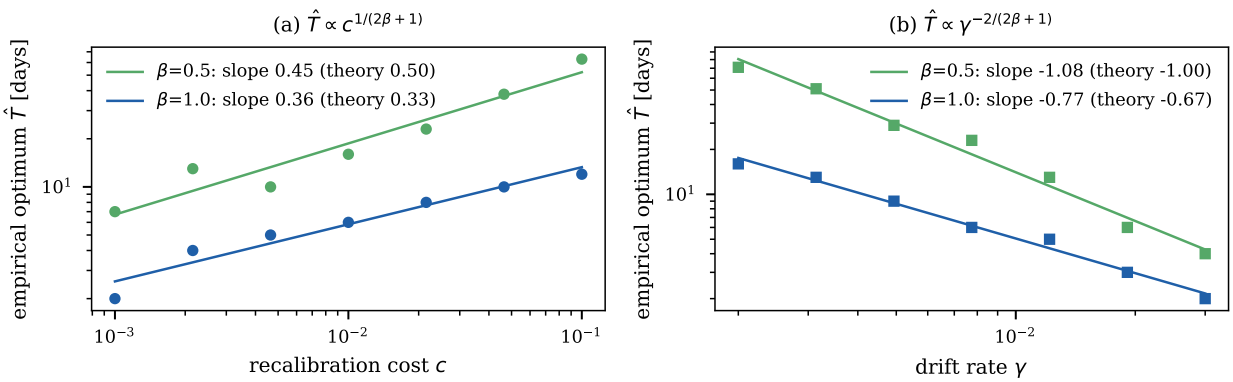

Fig. 2. Empirical optimum $\hat{T}$ (Monte-Carlo argmin, integer days) against (a) recalibration cost and (b) drift rate, on log–log axes, for $\beta = 0.5$ and $\beta = 1$. Fitted slopes bracket the theoretical exponents $1/(2\beta + 1)$ and $-2/(2\beta + 1)$ of Corollary 2.

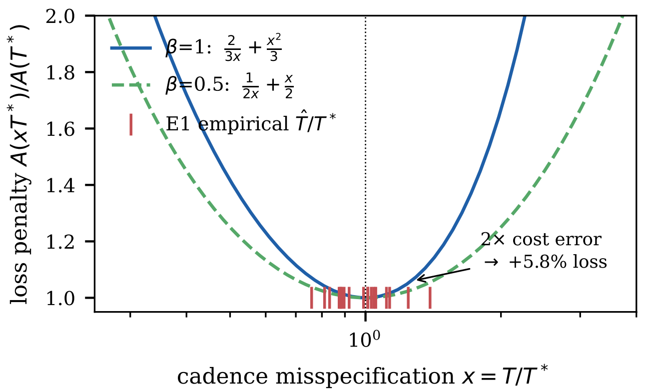

Fig. 3. Closed-form misspecification penalty (4) for $\beta = 1$ and $\beta = 0.5$, with the 18 E1 ratios $\hat{T}/T^*$ (ticks at R=1). The curve is flat near the optimum: a 2× cost misestimate costs 5.8% at β=1.

more than the law charges, so the prescribed cadence remains an upper bound on the optimal interval.

C. *The law, applied*

Pricing one recalibration at $c = 0.01$ Brier-day equivalents (the loss of tolerating a 10 pp gap for one day; the system's actual compute cost is seconds, so c is dominated by validation and operational risk) and taking the measured $\hat{\gamma} = 0.80$ pp/day:

- *Cube-root law* (β=1, E_0=0): $T^* = (3c/2\gamma^2)^{1/3} = 6.2$ days.
- *Affine law* (measured $E_0 = 14.8$ pp, Theorem 2): the cubic (5) gives $T^* \approx 2.8$ days — the residual baseline gap nearly halves the optimal interval.
- *Diffusive reading* (β=$\frac{1}{2}$, fitted $\gamma = 7.5$ pp·day$^{-1/2}$): $T^* = (2c/\gamma^2)^{1/2} \approx 1.9$ days.

Every reading lands in single-digit days: under measured drift, the common monthly or quarterly recalibration cadence is an order of magnitude too slow ($R(x)$ at $x \approx 30/6.2$ with β=1: a $\sim 8\times$ loss penalty), while the deployed system's refit-at-every-issuance policy is near-optimal precisely because its marginal recalibration cost is small.

VII. Discussion and Limitations

What the law is. A first-order design rule with the same epistemic status as EOQ [17]: exact under stylized assumptions, flat near its optimum (4), and useful chiefly because it is *legible* — it tells an MLOps team which way the cadence moves when compute prices fall ($T^* \propto c^{1/(2\beta+1)}$) or when a regime change doubles drift ($T^* \propto \gamma^{-2/(2\beta+1)}$). Reactive monitors [13]–[15] remain the right tool for catching discrete breaks; the law sets the baseline clock between alarms.

Cost commensuration. c must be priced in loss units, the law's most practitioner-dependent input. Corollary 3 is the mitigation: cost errors are attenuated to the $1/(2\beta+1)$ power, so even a 2× pricing error moves loss by 5.8% at β=1.

Identification caveats. Our ledger conflates staleness with lead time by design; disentangling them requires issuing forecasts of fixed lead from models of varying age, which we leave to future ledger accumulation. Calendar days (NSE trades $\sim$5/7) and a three-month, six-ticker, single-market window further widen the intervals; $\hat{\gamma}$ is triangulated across two channels but its CI includes values that would move T^* by a factor of $\sim$2 — which, per (4), costs under 26% in loss. Label latency (outcomes resolve k days late) delays $\hat{\gamma}$ updates but does not bias them.

When the model is also losing discrimination (our Fig. 4b), recalibration alone cannot restore performance, and T^* should be read as the inner loop of a two-clock policy: recalibrate on T^*, retrain on the slower clock that the retraining-economics literature [13], [16] addresses. Deriving the joint two-clock optimum in closed form is the natural next step, as is replacing Assumption 1 with a stochastic gap process (the β=$\frac{1}{2}$ law is the RMS shadow of exactly such a process) and anchoring the ledger cryptographically so that $\hat{\gamma}$ is auditable by third parties.

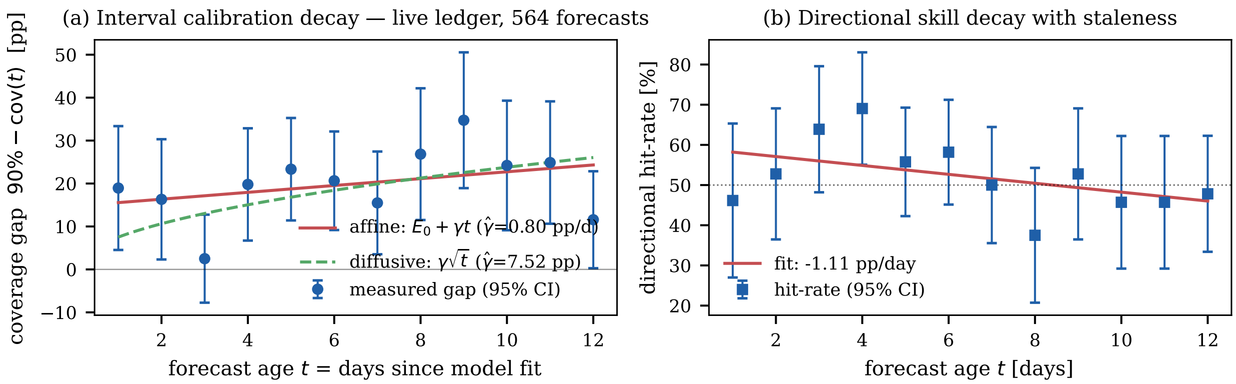

Fig. 4. Live-ledger calibration decay (564 resolved NSE forecasts, 2026-04-19 to 2026-06-11). (a) Gap between nominal 90% and empirical coverage by staleness, with affine and diffusive fits. (b) Directional hit-rate by staleness. Error bars are binomial 95% CIs.

VIII. Conclusion

We derived a closed-form law for the recalibration cadence of deployed probabilistic forecasters: $T^* = [(2\beta + 1)c/(2\beta\rho\gamma^2)]^{1/(2\beta+1)}$, bridging measured calibration decay to recoverable Brier loss through an elementary reliability bound, and inheriting the renewal–reward structure of classical maintenance theory. The law's exponents, optimum, and misspecification penalty are all analytic; simulations confirm each; and a live trading ledger supplies the first measured calibration drift rate we are aware of for a deployed financial forecaster — ≈ 0.8 pp/day, enough to make weekly-or-faster recalibration optimal at realistic costs. The broader argument is methodological: calibration maintenance deserves laws, not only monitors, and the century-old toolkit of operations research transfers to probability calibration as soon as one expresses drift in the right currency — the reliability term of a proper scoring rule.

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